Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 1 of 5 · Question 30 part whole · 2 marks

2024 JUN · Paper 02 · find part whole in this mark scheme extract

Question number	Answer	Mark
30	Up to **two** marks for understanding the influence of the pineal gland and melatonin on a person's sleep. For example: • The pineal gland activates/deactivates when it receives information about levels of light (from the external environment) (1) Melatonin is then released by the pineal gland which makes a person feel sleepy (1) Accept any other appropriate response.	**(2)**

Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 2 of 5 · Question 31 part whole · 2 marks

2020 NOV · Paper 02 · find part whole in this mark scheme extract

Question number	Answer	Mark
31	**One** mark for accurate understanding of pineal gland. **One** mark for exemplification of how pineal gland can be used to explain the scenario. For example: • The pineal gland regulates the secretion of melatonin so when it is light melatonin will reduce (1), so Pragya struggles to wake and is too tired and sleepy because her pineal gland is not sufficiently reducing her melatonin level enough to stay awake (1). Accept any other appropriate response.	**(2)**

Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 3 of 5 · Question 34 part whole · 4 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Question number	Answer	Mark
34	**One** mark for identification of each way (maximum **two** marks). **One** mark for justification which must be linked to the way identified (maximum **two** marks). For example: • The bright screen of the mobile phone could be a bright enough light source to result in Paolo struggling to sleep due to limited release of melatonin (1) as the light into the retina would prevent the suprachiasmatic nucleus (SCN) from triggering the pineal gland, so Paolo would not feel sleepy (1). • When Paolo stopped using the mobile phone and made his house darker he was able to sleep due to the increase of melatonin into his blood stream (1) because the pineal gland would have been activated to produce the sleep hormone as a result of the darkness, making Paolo less alert (1). Accept any other appropriate response.	**(4)**

Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 3 of 5 · Question 34 part whole · 4 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 4 of 5 · Question 33 part whole · 9 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

Question number	Indicative content	Mark
33	AO1 = 3 marks; AO2 = 3 marks; AO3 = 3 marks Markers must apply the descriptors in line with the general marking guidance (on pages 2–3). Due to the intrinsic links between the skills, if a response evidences performance against only one trait/skill it will be for demonstrating knowledge and understanding. An answer displaying qualities of AO1 only, cannot be awarded more than the top of Level 1, no matter how strong the performance is in AO1. AO1 • The pineal gland is located in the brain and is responsible for the production of melatonin. • Melatonin is a hormone that rises in the evening when it begins to go dark to prepare a person for sleeping. • Hormones help regulate the circadian rhythm which is an internal body clock that helps maintain the sleep-wake cycle. AO2 • The doctor may test the functioning of Basil's pineal gland to see if the suprachiasmatic nucleus is signalling to produce or reduce sleep hormones. • Basil may not be exposed to sufficient light in the day so is not then producing sufficient melatonin at night to make him sleepy and finds it hard to fall asleep. • The medication may not be fully working because of external factors such as insufficient light in the morning that may also cause Basil to have problems waking up. AO3 • Levels of hormones and pineal gland functioning can be tested empirically using scientific methods increasing the scientific credibility of hormonal explanations of Basil's sleep difficulty. • Evidence from Siffre (1975) suggests that zeitgebers have an influential role as exogenous pacemakers in setting the internal body clock for the sleep-wake cycle, so hormones may not be the only influence on Basil. • Skene and Arendt (2007) showed that blind people with some light perception have normal circadian rhythms but those without light perception show abnormal circadian rhythms, so Basil could also need light simulation to help regulate his circadian rhythm.	**(9)**

Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 4 of 5 · Question 33 part whole · 9 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding of a limited range of psychological ideas (AO1). • Provides little or no reference to relevant psychological ideas related to the context (AO2). • Limited attempt to deconstruct relevant psychological ideas. Limited consideration of supporting/refuting evidence, leading to generic judgements (AO3).
Level 2	4–6	• Demonstrates mostly accurate understanding of some relevant psychological ideas (AO1). • Provides some reference to relevant psychological ideas related to the context but this may be limited or lack relevance at times (AO2). • Deconstructs relevant psychological ideas using mostly logical chains of reasoning. Some consideration of supporting/refuting evidence, leading to a judgement (AO3).
Level 3	7–9	• Demonstrates accurate and thorough knowledge and understanding of relevant psychological ideas (AO1). • Provides sustained reference to relevant psychological ideas related to the context (AO2). • Deconstructs relevant psychological ideas using logical chains of reasoning. Sustained consideration of supporting/refuting evidence, showing an awareness of competing arguments, leading to a judgement (AO3).

Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 4 of 5 · Question 33 part whole · 9 marks

2022 JUN · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Section F
Language, thought and communication – How do you communicate with others?

Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 5 of 5 · Question 32 part whole · 4 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Question number	Answer	Mark
32	**One** mark for identification of strength/weakness (maximum **two** marks). **One** mark for justification, which must be linked to the strength/weakness identified (maximum **two** marks). For example: Strength • Tilly can have scientific medical tests for a low production of melatonin that could be making her unable go to sleep at night and staying awake when she should be sleeping (1), which gives the hormonal explanation of sleep problems credibility as hormones can be empirically tested to measure the levels when someone like Tilly is struggling to sleep (1). Weakness • There may be other factors causing Tilly not to be able to sleep as she could be worrying about her schoolwork so much that it is causing her to be unable to go to sleep at night (1) which would mean that considering just her hormone levels ignores any other factors such as stress that could be causing the problems as it focuses too much on biological factors and ignores environmental factors (1). Accept any other appropriate response.	**(4)**

Type MS · Hormones and sleep: the pineal gland and melatonin

Answer 5 of 5 · Question 32 part whole · 4 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract